

Express Riddler

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Riddle:

You are on the 10th floor of a tower and want to exit on the first floor. You get into the elevator and hit 1. However, this elevator is malfunctioning in a specific way. When you hit 1, it correctly registers the request to descend, but it randomly selects some floor below your current floor (including the first floor). The car then stops at that floor. If it's not the first floor, you again hit 1 and the process repeats.

Assuming you are the only passenger on the elevator, how many floors on average will it stop at (including your final stop, the first floor) until you exit?

Solution:

I let E_n be the expected number of stops to get from the $(n+1)^{\text{th}}$ floor to the first floor. So, getting from the tenth floor to the first floor would require E_9 stops on average.

Trivially, $E_0 = 0$ and $E_1 = 1$, because starting from the second floor, there is only the possibility of stopping on the first floor after a single stop. Starting from the $(n+1)^{\text{th}}$ floor, it takes a single stop to get to some new floor $k+1$, and from that first stop, the expected number of additional stops is E_k for $k < n$. For each floor below $n+1$, there is a $1/n$ chance of the first stop being that floor. So E_n can be written recursively as

$$\begin{aligned} E_n &= 1 + \frac{1}{n}(E_0 + \cdots + E_{n-1}) \\ &= 1 + \frac{1}{n}(E_1 + \cdots + E_{n-1}) \\ &= 1 + \frac{1}{n} \sum_{k=1}^{n-1} E_k \end{aligned}$$

Rearranging and adjusting the index of this equation gives

$$\sum_{k=1}^n E_k = (n+1) \left(E_{n+1} - \frac{1}{n+1} \right) - n$$

The harmonic numbers are defined by

$$H_{n+1} = H_n + \frac{1}{n+1}$$

with $H_1 = 1$. Another known identity of the harmonics numbers is

$$\begin{aligned} \sum_{k=1}^n H_k &= (n+1)H_n - n \\ &= (n+1) \left(H_{n+1} - \frac{1}{n+1} \right) - n \end{aligned}$$

Thus, H_n and E_n are defined identically, so $E_n = H_n$. The ninth harmonic number H_9 is $7129/2520 \approx 2.829$, so the solution is **7129/2520**.